

POWER SYSTEM REDUCTION TECHNIQUES FOR DIRECT TRANSIENT STABILITY METHODS

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Abstract. To speed up transient stability assessment of direct methods we use power system reduction techniques, combining the composite electromechanical distance (CED) approach with node reduction : the CED approach to identify and ultimately aggregate machines (almost) irrelevant to the transient phenomena; node reduction to eliminate irrelevant nodes, while possibly taking advantage of techniques such as sparsity preserving and connectivity based ones. The resulting approach is illustrated on the French EHV power system. It uses direct methods of the extended equal area criterion type, together with a CED appropriately tailored to the system specifics and with various node reduction techniques. Many variants so devised are compared. It is found that the CED approach contributes to significantly speed up the direct methods while fully preserving accuracy.

Keywords. Transient stability; direct methods; extended equal area criterion; composite electromechanical distances; external equivalents; power system reduction.

1. INTRODUCTION

Transient stability assessment requires the systematic running of multitudinous cases. In the particular contexts of short-term operation planning and real-time operation, where time is crucial, the need for fast methods becomes imperative. During the past three decades, numerous Lyapunov type direct methods have been widely investigated to meet this need to the extent possible [1]. They essentially amount to avoiding the stability time-domain computations of the system in its post-fault configuration.

The Extended Equal Area Criterion (EEAC) recently developed seems to be an interesting alternative : it is free from all time-domain computations, hence extremely fast [2 to 5]. Nevertheless, it can be rendered even faster by coupling it with reduction techniques [6]. The efficiency is normally expected to increase with the power system size, i.e. with the number of both machines and nodes that may be reduced.

To reduce the number of machines, we use the "Composite Electromechanical Distance" (CED) method [7, 8] belonging to the general category of "distance" approaches [9, 10]. On the basis of a CED adequately defined, this method classifies the machines in terms of their relevance to the stability phenomena of

concern, and identifies those which may be replaced by a simple equivalent without significantly distorting the phenomena. The derived reduced system may hold valid for a group of contingencies - and this can be interesting for multicontingency screening.

To reduce the number of nodes, we envisage various nodal elimination techniques, either of the conventional bus-by-bus ordered elimination, or of the sparsity preserving and connectivity based ones. Different such techniques examined in [11] are revisited here and explored more in-depth.

The resulting combined approach "CED along with nodal elimination" is illustrated by a systematic stability study conducted on the EDF system, using the EEAC. More specifically, the study started being conducted using the standard, basic EEAC [2 to 4]. In the meantime, i.e. quite recently, another, "dynamic" (as opposed to the basic, "static") EEAC has been developed with enhanced reliability performances [17, 18]. We thought it of interest to confront the coupling of reduction techniques with either of the above types of EEAC. The results obtained on the French network consistently show that the system aggregation via the CED approach fully preserves accuracy of the EEAC methods while providing important computing gains : compared with the time-domain numerical integration methods the proposed coupled "EEAC plus CED" reaches speeding up ratios of about 15 for single-contingency and 80 for multicontingency scenarios.

2. THRUST OF THE EEAC

Principle

For an assigned contingency, (i) decompose the multimachine system into the "cluster of critical machine(s)", which we henceforth refer to as the "critical cluster" (CC), and the group of the remaining machines; CC is presumably containing the machines which are responsible of the system loss of synchronism whenever it arises; (ii) reduce each group of machines into an equivalent machine, by "freezing" the individual rotor angles with respect to the corresponding partial center of angles; (iii) replace the equivalent two-machine system so obtained by a one-machine-infinite-bus (OMIB) system; (iv) apply the equal area criterion to OMIB, to assess the multimachine system stability via two measures : the stability margin corresponding to a given clearing time, and the critical clearing time (CCT) corresponding to a zero-valued margin. The use of appropriate Taylor series in conjunction with the OMIB succeeds in eliminating all step-by-step numerical integration.

A crux of the method is the CC identification. A current practice consists of choosing a number of candidate critical machines, of considering various candidate CCs (resulting from the combination of (some of) these machines, and of running the EEAC to

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determine the corresponding candidate CCTs; the actual CC is the one providing the minimum candidate CCT (since the EEAC provides sufficient and necessary stability conditions).

Basic, "static" EEAC [2 to 4]

One main feature of this "static-wise" approach, is that the machine rotor angles with respect to their respective center of angles are frozen once and for all, at the beginning of the transients. This, together with appropriate Taylor series, provides analytical close form expressions for both stability margin and CCT, and renders the stability runs extraordinarily fast.

On the other hand, candidate CCs are currently selected through the systematic combination of candidate critical machines taken one-by-one, two-by-two, etc.; for a number of, say, c candidate critical machines this yields $2^c - 1$ candidate CCs. A limitation of this "exhaustive combinatorial" is that in order to be tractable in practice, it has to be restricted to, say, $c \sim 9$ or 10 : hence, it cannot handle very large CCs; yet, such cases may exist, as e.g. in the French system, where CCs containing over 18 machines are not exceptional.

Candidate critical machines are selected on the basis of the "composite criterion" recently proposed [4], which improves the previously used "acceleration" one. In short, this criterion consists of relying on the machines' initial accelerations combined with their distance from the fault location in both pre- and post-fault configurations; these distances are computed via appropriately reduced admittance matrices.

Dynamic EEAC [17, 18]

It essentially pursues a twofold objective. (i) To improve accuracy, it uses a dynamic OMIB whose parameters are refreshed in both during- and post-fault configurations, instead of being kept frozen once and for all at the beginning of the transients, as in the static approach. (ii) To improve the CC identification, it selects the machines in the order they should be combined; this is achieved by computing the nearly critical trajectory of the individual machines and by considering their rotor angles at the time they nearly reach their unstable equilibrium [17, 18]. By doing so, one reduces the combination of c machines to c (candidate CCs).

From the viewpoint of reliability and accuracy, this dynamic EEAC is substantially improved both because it avoids inaccuracies inherent to the "static" OMIB and because it becomes able to handle any number of critical machines, whereas the "exhaustive combinatorial" of the basic approach could not.

From the viewpoint of computing requirements, running a candidate dynamic OMIB takes longer than a static one; but this is compensated by the dramatic decrease of the number of candidate CCs and hence OMIBs with respect to the static ones.

3. THE CED METHOD

Principle [7, 8]

For a given contingency and within the simplified system modelling, it decomposes the system machines into two groups:

- the internal group contains the machines deemed relevant to the phenomena of concern and therefore kept intact;
- the external group contains the machines which are "too far away" from the contingency, i.e. (almost) irrelevant. These machines will be aggregated into one or a few number of equivalent machines.

The decomposition relies on the use of a CED able to classify the machines in terms of their relevance to the stability phenomena. It results from the combination of elementary electromechanical distances appropriately chosen.

Elementary electromechanical distances

Reference [8] proposes and combines various such elementary distances, and further compares them in the context of a 31-machine system used for illustration. A similar exploration has been carried out for the French 61-machine system. Three elementary distances were found to provide the most efficient CED; for machine i ($i = 1, 2, \dots, n$, n being the total number of system machines) these are:

- the absolute value of the acceleration of machine i evaluated at the time immediately following the contingency inception;
- the modulus of the pre-fault transfer admittance between the faulted node f and the internal node of machine i , Y_{if}^0 ;
- the product $Y_{if}^0 * M_i$ where M_i is the inertia coefficient of machine i . Note that inertia coefficients have proved to be quite interesting "distances", provided that they are weighted by a "topological" or "electrical" distance factor; this holds true especially for large power systems.

Aggregated system of level m

This is the system resulting from the aggregation of the m "less relevant" machines, according to the CED classification. In relatively small power systems, these m machines may be aggregated into a single equivalent machine [12]. It is connected through a very small transient reactance to a fictive node obtained from the aggregation of the m machines' internal nodes; the reactance is introduced to comply with the standard simplified machine model.

For large power systems, significant gains may be obtained (i.e. more machines may be aggregated without loss of accuracy) by considering more than one equivalent machines, corresponding to a geographical or electrical decomposition of the system. For the French system, the decomposition is described in section 5.

Off- and on-line determination of the external machines

The off-line determination of the external machines of an n -machine system consists of considering successively the aggregated systems of levels $n-1$, $n-2$, ..., 2, assessing their accuracy, and choosing an "optimal" aggregation level.

The accuracy assessment of each aggregated system relies on two complementary evaluation criteria: the mean absolute value (MEA) and the maximum absolute value (MAA) of the rotor angle error between the original and the aggregated systems [8]. They are obtained from a time-domain numerical simulation of the contingency. Figure 2 of section 5 illustrates MEA curves plotted versus aggregation level m .

The on-line determination of the external machines relies on the observation that to each aggregation level corresponds a CED value [8, 12]. Hence, repeating the above procedure for different operating conditions, allows to associate a threshold CED value, V_{th} , to the given contingency. This yields the following on-line procedure: for given contingency and operating conditions, (i) compute the CED and its values, (ii) consider that the machines having a CED value $< V_{th}$ belong to the external group and can be aggregated.

Extensions

The CED method can be extended to the identification of an aggregated system valid for a zone of contingency locations instead of a sole one. This "contingency zone" (CZ) relies on the concept of correlation between contingencies [8, 12].

Similarly, for a given contingency, the correlation between two operating conditions may easily be defined, allowing one to derive aggregated systems valid for a set of operating conditions.

As a result, off-line studies allow to identify CZs and the corresponding external groups of machines, valid for a range of op-

erating conditions. For each new case, the correlation coefficient provides an on-line measure of the goodness of the aggregated systems obtained off-line.

Practical use of the CED

The CED approach identifies the aggregated system of level m valid for a CZ defined with respect to a given "central" contingency location. Let us specify the outcomes of the approach.

(i) Aggregated system of level m : if n is the total number of system machines, and if the m machines are aggregated into, say e equivalent ones, then the number of machines kept individual is $I = n - m$, and the total number of machines of the resulting equivalent system is $n_{res} = I + e$. To fix ideas, in the case study of section 5, $n = 61$, $m = 30$; $e = 4$, $n_{res} = 35$.

Unless otherwise specified, in the sequel, we shall refer to all n_{res} machines of the aggregated system simply as the "machines".

(ii) CZ and its composition : CZ is composed of external nodes of properly defined individual machines, belonging to the I above considered ones, and of load nodes. To fix ideas, the CZ of the following case study comprises 5 machine nodes and 15 load nodes.

4. NODE REDUCTION

Problem statement

From the original unreduced system we derive an equivalent one, which we shall henceforth refer to as the "reduced system". Its nodes are at the maximum of the following two categories.¹

- The "mandatory nodes". These are : (i) the machines internal nodes²; (ii) the nodes belonging to the CZ, identified by the CED method; (iii) those adjacent to the above CZ nodes, and hence necessary to the stability scenarios : namely, because the post-fault configuration may imply the opening of lines adjacent to the node where the fault applies.

- The nodes retained in order to enhance the efficiency of the reduced system or to satisfy other desired properties, dictated by sparsity and/or connectivity concerns discussed below.

All nodes not belonging to the reduced system ones are labelled "external nodes". They will subsequently be eliminated by means of e.g. a conventional bus-by-bus ordered elimination.

Sparsity-based technique

It aims at enhancing or preserving to the extent possible the sparse characteristics of the reduced system. In our particular context, we will explore possible advantages of the sparsity preserving technique denoted hereafter SP, and based on the so-called "ordering scheme 2" [13].

This technique, used also in [14], derives from a general algorithm proposed by Tinney and Peterson [15]. Applied to our problem, it has the following main features.

- At the beginning of the process, all the non-mandatory nodes start being eliminated according to the specified node ordering scheme 2.

- In the course of this process, each time a node is eliminated, the algorithm computes the number B of branches in the reduced system formed by the remaining nodes.

- Examining the evolution of B , the algorithm finds the optimum of a certain sparsity criterion.

¹In the sequel, we assume that a CZ has been identified by the CED method and that the corresponding external machines have been aggregated into the appropriate equivalent ones, so that the system comprises $I + e$ machines.

²Remember, these are required by the EEAC, which involves computation of the admittance matrix reduced at the machines internal nodes.

The nodes not eliminated at the optimum belong to the reduced system.

The simplest sparsity criterion consists of minimizing B . However, in the EEAC method, the number N of nodes in the reduced system is of importance. Therefore, as proposed by Tinney and co-authors [16], we also investigate the weighted sum $N + 2B$.

Note that another technique based on iterative partial eliminations proposed in [11] has also potential.

Connectivity-based technique

It aims at identifying nodes additional to the mandatory ones, to form a connected internal area in the original system. Indeed, it may appear sound that every pair of nodes taking an active part in the transient stability phenomena are connected, in the original system, by paths entirely included in the internal area.

We name "path", an ordered set of nodes such that each node is connected to its successor by a system branch (line or transformer). In the beginning of the process, the internal area contains only the mandatory nodes. The proposed algorithm consists of two parts.

The first scans this internal area and identifies its largest "islands", i.e. the largest sets of mandatory nodes connected by paths entirely included in it. The following step appends the machines external nodes. Then the method searches for possible "single-node" paths between islands; such a path is composed of one external node adjacent to at least two nodes belonging to different islands. When a "single-node" path is found, the corresponding external node is included in the internal area, and the islands connected by this path are merged into a larger one.

In the second part, the algorithm searches for the shortest paths connecting the remaining islands. The procedure relies on graph theory and makes use of Dijkstra's algorithm. Note that the length of a path may be evaluated in various ways, in particular the following two :

- a topological length, expressed in terms of the number of nodes in the path; this yields the so-called "topological variant" of the connectivity-based technique;
- an electrical length, expressed as the sum of the moduli of the branch impedances in the path; this yields the "electrical variant".

Combining connectivity- and sparsity-based techniques

Combining the two types of approaches leads to a compromise : on the one hand, achieving the connectivity of the internal area may spoil the sparsity of the reduced system; on the other, the additional nodes identified on the basis of sparsity considerations may form new islands. However, these latter nodes do not really take an active part in the transient stability phenomena and therefore need not form a connected area with the other internal nodes in the original system. Consequently, in this paper, we shall apply first the connectivity-based technique (with electrical or topological paths), then the sparsity-based technique.

5. SIMULATION CONDITIONS

System description and decomposition

The considered system is an outdated version of the 400-kV and 225-kV EDF power system, comprising 61 machines (of which some are equivalent representations of the surrounding European interconnections), 561 buses, 810 lines and 190 transformers. In this first stage study, only one operating condition has been considered.

Prior to the application of the CED approach per se, the system has been decomposed into 4 main regions. To this end, the admittance matrix reduced at the machines internal nodes was

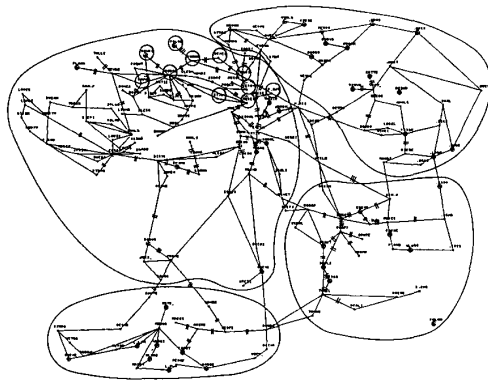


Figure 1. Schematic representation of the EDF system and its 4 main regions

computed, and the moduli of the transfer admittances between every pair of machines have been listed in a decreasing order of their values. Scanning the terms above a specified threshold (0.6 p.u.) yields a decomposition of all the machines but five into 4 main regions. The remaining five machines, on the basis of the largest terms relating them to other machines, were incorporated in the regions to which these latter machines belong. The regions so identified correspond in fact to the 4 distinct geographical areas illustrated in the schematic system representation of Fig. 1, namely :

- a North-East region near Belgium and Germany comprising 18 machines;
- a South-East region near Switzerland and mainly corresponding to the Rhône valley, comprising 14 machines;
- a South-West region near Spain with 8 machines;
- a region covering the North-West and the center with 21 machines.

Contingency zone (CZ)

In this study, focus has been placed on the "Paluel area", located in the North-West part of the network, and on contingencies applied at "nearby" machines and "nearby" load nodes of the 400-kV level. The centre of the area has been considered to be "Bar". To specify the "nearby" Bar machine and load nodes, use has been made of the CED procedure developed in Ref.[8]. In short, this has consisted of the following steps.

(i) Taking "Bar" as the centre fault location, classify the machines according to the CED described in Section 3; consider the top machines of this list and compute their respective correlation coefficient with respect to "Bar"; select those machines whose coefficients are above 0.95, taken as the threshold. This step has provided machines P1, P2, P3, P4, H4 as the "zonal" machines, for which a zonal CED has been computed.

(ii) To fully determine the contingency zone, the load nodes must also be specified. To this, a procedure similar to the above is followed where, however, the zonal CED is used instead of the CED relative to "Bar"; the same as above threshold of 0.95 is used to delimit those nodes which belong to the zone. This leads to selecting 15 local nodes.

(iii) Accordingly, the CZ comprises 20 nodes; actually, they correspond to 11 different substations represented by small circles in Fig. 1.

It is interesting to mention that the above CZ is in a very good agreement with the "study region" determined by EDF engineers in charge of stability studies relating to the above CZ center node.

To determine the appropriate aggregation level for the CZ,

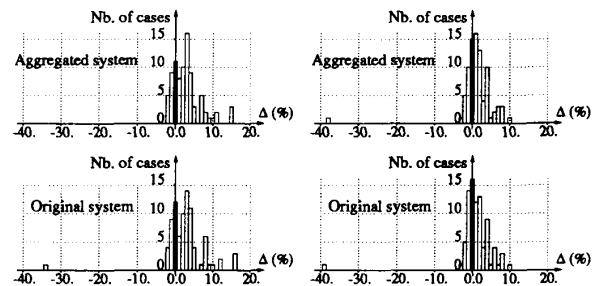


Figure 3. Static EEAC error distribution

Figure 4. Dynamic EEAC error distribution

MEA and MAA curves have been drawn for the 20 CZ nodes. Two typical samples of MEA curves are shown in Fig. 2 : one corresponds to the center of the CZ, Bar, the other to a load node nearby the CZ border. Considering an accuracy such that $MEA \leq 3^\circ$ and $MAA \leq 10^\circ$ for all CZ nodes, leads to an aggregation level of 30 for the whole CZ.

Hence, the 30 last machines of the classification (derived for the CZ) are aggregated into 4 equivalent machines according to the system decomposition into 4 regions. This yields a system of 35 machines equivalent to the original one; and this will be our "aggregated system of level 30" or simply "aggregated system", as opposed to the original "unaggregated" one of 61 machines.

Simulated contingencies

The type of considered contingencies is three-phase short-circuits applied at each one of the above 20 CZ nodes. They concern three different scenarios : "single-line" faults where the fault is cleared by opening a single line adjacent to the faulted node, "double-line" faults where the fault is cleared by opening a double line ending up at the faulted node, and "busbar" faults where the fault is cleared by opening all the elements (lines and transformers) ending up at the faulted node. They provide respectively 51, 22 and 12 contingencies, totalling 85.

Accuracy assessment

Hereafter we compare the results provided by the application of the EEAC (static as well as dynamic) on the aggregated system (symbolically denoted EEAC(agg)) and on the original system (denoted EEAC(orig)). Note that the node reduction, performed according to any of the schemes presented in section 4, does not affect the accuracy. This latter depends only on the machine aggregation.

Comparison of CCTs. The "error" of the CCT(EEAC) with respect to the CCT(SBS), where the step-by-step (SBS) numerical integration method is taken as the benchmark, is expressed by

$$\Delta = \frac{CCT(SBS) - CCT(EEAC)}{CCT(SBS)} \times 100 \quad (\%)$$

The error distributions on the 85 contingencies for the original and the aggregated systems are compared in Fig. 3 for the static and in Fig. 4 for the dynamic EEAC. Obviously, the corresponding histograms are very similar; a closer examination shows that the largest deterioration of the aggregated vis-a-vis the original system is of 0.64 % for the static and 0.27 % for the dynamic EEAC! Incidentally, we note that the dynamic EEAC is more accurate than the static one which, however, is quite accurate, too.

Comparison of critical clusters (CCs). In both static and dynamic EEAC, the CCs are composed of some of (and sometimes all) the 7 machines P1 P2 P3 P4 F1 F2 and H4. For the dynamic EEAC the CCs provided by EEAC(orig) and EEAC(agg) are

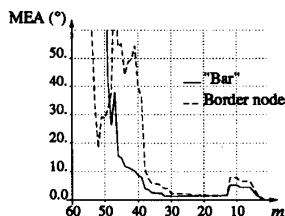


Figure 2. Two MEA curves

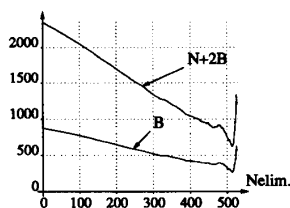


Figure 5. Application of SP

exactly the same; for the static EEAC they differ in 8 cases, but this hardly affects the CCT accuracy; indeed, these differences proceed from a mere inversion of the first two candidate CCs (very close in terms of CCTs), probably due to very small numerical inaccuracies.

Note that the mean number of candidate CCs considered by the static EEAC varies in between 1 and 495; over the 85 contingencies this yields a mean number of 265. The mean number of candidate CCs for the dynamic EEAC is 15.

6. COMPUTING TIMES

We assess and compare the computational gains resulting from the coupling of the CED and the various proposed node reduction techniques with the EEAC. The results are reported in the parts of Tables 1 and 2, labelled "with CED". They are described below, along with the corresponding abbreviations used.

Description of the combinations used in Tables 1 and/or 2

Original system (orig): concerns the application of the EEAC to the original unaggregated, unreduced system.

Aggregation alone (aggreg): concerns the application of the EEAC to the aggregated system *without other node reduction*, i.e. to the equivalent system composed of 35 machines, *all load buses being kept intact*.

Mandatory nodes (MN): they are composed of the 20 nodes belonging to the contingency zone, 18 nodes adjacent to the above ones, and 35 machines' internal nodes for the aggregated system, totalling 73 nodes. The node reduction thus eliminates $(561 + 4 + 35 - 73) = 527$ nodes, using a conventional bus-by-bus ordered elimination scheme. Note that the number B of branches in the admittance matrix after the reduction is of 567, therefore $N + 2B = 1207$.

Besides MN, the node reduction techniques reported hereafter identify additional nodes which are also retained in the reduced system. Table 1 gives for each one of them the number AD of additional nodes, the numbers N of nodes and B of branches in the reduced system and the value of the sparsity criterion $N + 2B$.

Sparsity-based technique (SP). Figure 5 illustrates the application of SP on the aggregated system (i.e. column SP of the "with CED" parts of the tables). The two curves show the evolution of the sparsity criteria B and $N + 2B$ with the number of eliminated non-mandatory nodes. For $N + 2B$, the minimum is reached for $N_{elim} = 514$ and $N + 2B = 620$; this minimum also corresponds to the minimum of B ($B = 267$).

Connectivity-based techniques. The two variants corresponding to electrical and topological paths are considered and are respectively denoted EL and TOP.

Combination of connectivity- and sparsity-based techniques. The combinations of SP with EL and TOP are considered. In both cases, the minimum of $N + 2B$ corresponds to the minimum of B , their values are indicated in the left hand part of Table 1. The resulting numbers of nodes are also given.

Table 1 - Additional nodes and sparsity criteria

	With CED						Without CED					
	MN	SP	EL	TOP	EL+SP	TOP+SP	MN	SP	EL	TOP	EL+SP	TOP+SP
AD	0	13	57	54	59	57	0	60	104	103	127	128
N	73	86	130	127	132	130	99	159	203	202	226	227
B	567	267	293	367	283	289	1724	403	774	763	430	434
N + 2B	1207	620	716	861	698	708	3547	965	1751	1728	1086	1095

Computing steps

The EEAC computation process may be divided into 5 or 4 main steps depending upon whether power system reduction techniques are applied or not. These steps are described below and the corresponding computing times are reported in Table 2 for the different considered combinations. All times are given in seconds except in row 12. They are obtained on a SUN SPARC Station 1 of 12.5 MIPS, 1.4 DPMFLOPS and 16 MB of central memory.

Step 1 (row 1): loading the system data and operating condition, translating them into an internal format more appropriate for the computations, determining the machines electromotive forces behind transient reactances and converting the loads into constant impedances; it also includes the computation of the pre-fault unreduced admittance matrix. This step takes about 8.9 s and is the same whatever the method used. Therefore no special care has been taken to optimize it, and it could certainly be improved.

Step 2 (rows 2 to 5): derivation of the reduced system whenever needed. It is composed of 3 operations: the aggregation of the external machines (row 2), a load flow performed on the aggregated system to correct the initial conditions as a result of the machine aggregation (row 3), and the node reduction (row 4). Row 5 totals the times of these 3 operations.

Step 3 (row 6): reduction of the pre-fault admittance matrix of the system to the machines internal nodes (needed by the EEAC). Node ordering scheme 2 is used during the node elimination process. Step 3 as well as steps 1 and 2 are performed only once for a given operating point and are the same for the static and the dynamic EEACs.

Step 4 (row 7): computes the during-fault and post-fault admittance matrices reduced at the machines internal nodes. For the static EEAC, it also computes the distances required by the composite criterion (see Section 2) and thus implies an additional matrix reduction. Since the corresponding configurations are not much different from the pre-fault one, the matrix reductions of step 4 use the node order determined in step 3.

Step 5 (row 8): concerns the computations specific to the (static or dynamic) EEAC (see Section 2).

Steps 4 and 5 are performed once per contingency and the times reported in rows 7 and 8 result from the means over the 85 simulated contingencies. The values in row 8 correspond to the mean number of candidate CCs.

Row 9 sums up the times of all 5 steps to provide the average computing times for the single contingency case. However, since step 1 is the same whatever the method, the values in row 10 (sum of the last 4 steps) are more meaningful. For the multicontingency case, row 11 reports the average computing times required by each new contingency. Finally, the total times needed to simulate the 85 contingencies are given in minutes in row 12.

Discussion

Static EEAC: examination of the left hand part of Table 2 yields the following comments.

A first general observation stems from the comparison of column "orig" with those of "with CED". It shows that the machine aggregation has proportionally the greatest influence on the global

Table 2 - Computing times

		Static EEAC														Dynamic EEAC		
		With CED								Without CED						With CED		
		orig	aggreg	MN	SP	EL	TOP	EL+SP	TOP+SP	MN	SP	EL	TOP	EL+SP	TOP+SP	orig	aggreg	MN
1	step 1	8.9				8.9							8.9			8.9		
2	aggregation	0				0.06							0			0		0.06
3	load flow	0				5.3							0			0		5.3
4	node reduc.	0	0	0.65	1.61	1.11	0.99	1.93	1.89	0.98	2.03	0.76	0.80	1.39	1.46	0	0	0.65
5	step 2	0	5.36	6.01	6.97	6.47	6.35	7.29	7.25	0.98	2.03	0.76	0.80	1.39	1.46	0	5.36	6.01
6	step 3	1.68	1.03	0.29	0.34	0.43	0.49	0.46	0.45	0.91	1.15	1.65	1.66	1.22	1.20	1.68	1.03	0.29
7	step 4	3.82	1.90	0.85	1.00	1.26	1.45	1.34	1.32	2.75	3.34	4.81	4.84	3.59	3.49	2.62	1.33	0.60
8	step 5	8.07				2.75							8.07			10.65		3.85
9	5 steps	22.47	19.94	18.80	19.96	19.81	19.94	20.74	20.67	21.61	23.49	24.19	24.27	23.17	23.12	23.81	20.42	19.66
10	(9)−(1)	13.57	11.04	9.9	11.06	10.91	11.04	11.84	11.77	12.71	14.59	15.29	15.37	14.27	14.22	14.91	11.52	10.76
11	(7)+(8)	11.89	4.65	3.6	3.75	4.01	4.20	4.09	4.07	10.82	11.41	12.88	12.91	11.66	11.56	13.23	5.13	4.46
12	85 cases (min)	17.0	6.8	5.4	5.6	5.9	6.2	6.1	6.0	15.5	16.4	18.4	18.5	16.7	16.6	18.9	7.5	6.6

computing times : it divides the time for step 5 by almost 3 but also proves to significantly alleviate steps 3 and 4. These benefits largely compensate for the derivation of the reduced system (which takes between 5.36 s and 7.29 s). Note that the time needed to run the load flow has not been optimized and could certainly be improved.

More specific observations are related to the columns of the "with CED" part of the table.

- Applied alone, the machine aggregation (aggreg) shortens the times for step 3 and 4 with respect to the original system (orig), as well as the global times for the single contingency case (rows 9 and 10). In the multicontingency case, only 39% of the time is required per contingency (see in row 11). The simulation of the 85 contingencies takes 6.8 min instead of 17 min.

- However, it appears that the **best solution consists of simply reducing all nodes except for the mandatory ones (MN) in the aggregated system**. The times for steps 3 and 4 are substantially shortened with respect to "aggreg" (divided respectively by 3.6 and 2.2) and even more with respect to "orig" (division factors of 5.8 and 4.5 respectively). With respect to "aggreg", a global gain of more than 1 s is obtained per contingency (rows 9, 10 and 11); and the gain for the 85 contingencies is considerable with respect to "orig" (a factor 3.1 : 5.4 min versus 17 min). Note also that the derivation of the reduced system is relatively fast (6.01 s is the smallest time after "aggreg").

As a conclusion, this best solution succeeds in speeding up the multicontingency investigations of the EEAC by a factor 3.

- The sparsity technique SP slightly spoils the results obtained for MN ! The time required to derive the reduced system is large (the third time after EL+SP and TOP+SP). The computing times for steps 3 and 4 are divided by respectively 3 and 1.9 with respect to "aggreg", and 4.9 and 3.8 with respect to "orig". Comparing with "aggreg", no global saving is obtained for a single contingency, and the gain is of 0.9 s per contingency for the multicontingency case. Therefore, SP does not seem to be interesting for the problem at hand. (This is quite different from the step-by-step numerical integration method, where the computing time is a function of the number of terms in the factorized admittance matrix.)

- The connectivity variants EL and TOP yield negligible savings with respect to "aggreg"; they are certainly not competitive with MN.

- The combination of SP with EL and TOP does not seem to yield significant results (with respect to these latter, a small gain for TOP+SP for steps 3 and 4, no gain at all for EL+SP).

Remark : to globally appraise the advantages of the CED over the various node reduction techniques, we also applied these latter

directly on the original system (i.e. without the machine aggregation). The results are reported in the parts of Tables 1 and 2 labelled "without CED". The mandatory nodes are composed of the same 20 CZ nodes³ and 18 adjacent nodes, and of the 61 machines internal nodes, which makes 99 mandatory nodes. The node reductions only affect the computation of the reduced matrices (and of course the derivation of the reduced system). All in all, they yield only marginal global computational gains with respect to "orig", and sometimes no gain at all. More specifically :

- although the computing times to derive the reduced system are the smallest, both the electrical and topological connectivity variants EL and TOP give times larger than orig (except for step 3) ;
- the "best" node reduction appears to be the conventional bus-by-bus ordered elimination at the mandatory nodes (MN) : it speeds up steps 3 and 4, and yields a gain of almost 1 s per contingency (see rows 9, 10, 11) and of 1.5 min for the 85 cases.

- Again, the sparsity technique SP slightly spoils the results provided by MN (see the computing times for steps 3 and 4). No global saving is obtained for a single contingency, and very little (about 0.5 s per contingency) for the multicontingency case.

- The combination of SP with EL and TOP allows to improve the performances of the connectivity-based reduced systems. However, they are certainly not competitive with the one resulting from MN. Note that this improvement suggests that SP may be more efficient when the retained nodes are located differently in the system and when their number is larger (see Table 1).

Dynamic EEAC : the RH part of Table 2 reports only on the 3 more interesting combinations, namely "orig", "aggreg" and "MN". The computing times for the other combinations could approximately be obtained from the corresponding LH part of the table, since : (i) steps 1 to 3 are the same, (ii) step 4 is approximately 2/3 of the "static" one (as illustrated by the reported values) and (iii) step 5 is only affected by the machine aggregation.

As a consequence of points (i) and (ii), most observations concerning steps 3 and 4 still apply here. They are greatly alleviated by the machine aggregation; with respect to "orig", the division factors for step 4 are of the same order, namely of 2 for "aggreg" and of 4.4 for "MN".

Moreover, as already observed, the machine aggregation divides the time for step 5 by almost 3 and thus appears again as having the greatest influence on the computing times.

Examining rows 9 to 12 yields the following comments.

- "aggreg" : a gain of more than 3 s is obtained for the single contingency case (rows 9 and 10) whereas in the multicontingency case a gain of 61 % is achieved per contingency (row 11). For the

³still identified by the CED method

85 contingencies, this yields a time of 7.5 min versus 18.9 min.

• "MN" : again, "MN" appears to be the best solution. However the gain with respect to "aggreg" is slightly smaller than for the static EEAC : it amounts to 0.76 s for rows 9 and 10 (single-contingency case) and to 0.67 s for row 11 (multicontingency case). The simulation of the 85 contingencies takes 6.6 min versus 7.5 min. With respect to "orig", a factor of 2.9 is obtained (6.6 min versus 18.9 min).

Finally note that for both static and dynamic EEACs, step 5 appears to be proportionally the most time-consuming in the multicontingency case since it has to be multiplied by the number of contingencies; hence the interest in procedures which relieve it. The more step 5 will be alleviated, the faster will be the EEAC, and the less marginal will be the benefits brought by the node reduction.

Step-by-step numerical integration method

For comparisons, the CCTs for the 85 contingencies have also been evaluated on the same computer by the time-domain SBS numerical integration method with the following characteristics : integration period of 3 s (deemed necessary for accuracy); 4th order Runge-Kutta integration method with variable step size; CCT computed iteratively by dichotomy. For the considered contingencies, the starting search interval is [0, 1 s] and the procedure stops when the length of the search interval becomes less than 0.005 s : this results in 8 iterations per contingency.

The first part of the SBS process is the same as step 1 described above. Apart from this step performed only once per operating point and taking 8.9 s (see Table 2), the computation of one CCT requires on average 302 s (≈ 5 min). For the 85 contingencies, this totals to about 7 h 8 min; therefore, for the static EEAC the "best solution" of Table 2 (MN "with CED") is 79 times faster than the SBS method; for the dynamic EEAC, it is 65 times faster. For the single-contingency case, we obtain ratios of above 15. Observe that the EEAC algorithms are still of research type, where no particular care is taken to optimize their computational performances. The above ratios should be taken as indications rather than as reflecting the real EEAC performances, which could be easily improved.

7. CONCLUSION

Power system reduction techniques coupled with EEAC have been used to speed up transient stability assessment.

Two EEAC approaches have been used, viz. the standard, "static" and the new "dynamic" ones. Reduction techniques, on the other hand have combined the composite electromechanical distance (CED) approach with sparsity preserving and connectivity based node reduction approaches. The coupled "EEAC plus reduction" approaches have then been applied to the French EHV network. The main conclusions of this study are as follows. (i) The system machines aggregation via the CED fully preserves the accuracy of the EEAC method; even the largest recorded deterioration is negligible. (ii) Among the explored reduction techniques, the CED has shown to be - and by far - the determinative factor for computer gains. (iii) Comparing "EEAC + CED" vs. EEAC alone shows that CED contributes to speed up EEAC, especially in multicontingency cases where a speed up ratio of about 3 is obtained. (iv) The gains are considerably more impressive when comparing computing times of "EEAC + CED" with those of the step-by-step numerical integration with same simplified machine modelling : a first, rather conservative, pessimistic indication gives ratios of above 15 for single-contingency scenarios, and of about 80 for multicontingency (85) scenarios.

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